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Predictive Analytics Project: College Drop-out Memorandum

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To: Professor Callaghan, President of the College

From: ISDS 415 Project Group 7 Student Support Office

Subject: College Drop-out Analytics

Introduction

This report summarizes the findings of a predictive analytics study conducted by the Student Support Office at Mont St. Michel College. In recent years, the college has seen an increase in the number of first-year students who do not return for their second year. This trend highlights a need for more targeted institutional support. To address this concern, the college introduced voluntary one-credit seminars led by faculty, designed to help first year students develop stronger connections with the campus community. The purpose of this analysis is to determine whether those seminars have had a positive effect on student retention and identify the demographic and academic factors that best predict both seminar attendance and student drop-out.

A random sample of 600 first-year students from the latest academic year was analyzed. The data include each student's high school GPA, number of units enrolled, age, gender, registration status (part-time or full-time), commuter status, seminar attendance, and whether the student dropped out after their first year. Logistic Regression and MLP neural network were developed and assessed to predict seminar attendance and the likelihood of student drop-out. The report concludes with probability-based predictions for two outcomes and recommendations for college administration.

Descriptive Overview of the Sample

The 600 student sample represents the college's first year population. The high school GPA has a range of 1.93 to 4.00, with a mean of 2.90 and a median of 2.87 indicating a slight skewed distribution. Students enrolled in an average of 12.43 units per semester, with a median of 13 units, reflecting a full-time course load. Ages ranged from 18 to 43, with a mean of 21.42 and a median of 19. The histogram shows a strong right-skewed distribution that is consistent with the typical college age population. Of the sample of first-years, 52.8% identify as female and 47.2% as male. Most students were enrolled full-time (74.0%), while 26.0% were part-time. In addition, 68.3% of students lived on campus, while 37.7% commuted at least 20 miles to campus. Furthermore, 37.7% of participants attended the voluntary seminars while 62.3% did not. Overall, 20.0% of students dropped out after their first year.

While dropout rates were higher among seminar attendees (27.0%) than non-attendees (15.8%), this likely reflects selection bias, as academically at-risk students were more inclined to seek support. Students who dropped out had a lower average GPA (2.66 vs 2.97) and higher average units enrolled (13.53 vs 12.15) compared to those who remained. This is consistent with the original narrative that heavier course loads and lower GPA contribute to students dropping out. Notably, students who commute had a lower dropout rate (12.6%) than non-commuter students (23.4%). This suggests that commuter status is not a contributing factor to the increase in dropouts.

Descriptive Overview of the Sample

A logistic regression model was built to predict whether first-year students would attend the seminar. The model was statistically significant with an overall accuracy of around 66–67%, though it was better at identifying non-attendees than attendees.

Three variables stood out as significant predictors: high school GPA, age, and commuter status. Students with lower GPAs, older students, and commuters were all more likely to attend. Units enrolled, gender, and registration status did not show significant effects.

Seminar attendance isn't random for students who attend tend to have different characteristics than those who don't, which is worth keeping in mind when interpreting later results.

Predictive Models for Seminar Attendance

To evaluate the predictive performance of the seminar attendance model, a Multilayer Perceptron (MLP) neural network was applied using the independent variables. The dataset was split into training and testing samples, with approximately 70% used for training and 30% for testing.

The neural network achieved a training accuracy of 66.0% and a testing accuracy of 65.5%, as shown in the classification table. Compared to the logistic regression model, which had an accuracy of approximately 66.7%, the neural network performed similarly and did not provide a meaningful improvement in predictive performance.

Despite this slight improvement, logistic regression is preferred because it allows for clear interpretation of how each variable affects seminar attendance. The neural network, while useful for prediction, does not provide easily interpretable relationships between variables.

Predictive Models for Student Dropout

A logistic regression model was developed to predict the probability that students will drop out after the first year. The dependent variable was students who dropped out status, and all available independent variables were included.

The model was statistically significant and achieved an overall classification accuracy of approximately 80.8%. The model performed very well in predicting students who did not drop out, but it was less effective at identifying students who did drop out.

The results showed that high school GPA, units enrolled, commuter status, and seminars attended were statistically significant predictors. High school GPA had a strong negative relationship with dropouts, indicating that students with higher GPAs are less likely to drop out. Unit enrollment had a positive relationship, meaning that students taking more units are more likely to drop out. Commuter status was also significant. Seminar attendance showed a positive relationship with dropout, indicating that students who attended seminars were more likely to drop out.

However, this result should not be interpreted as seminars causing dropouts. Seminar attendance is influenced by student characteristics. Therefore, students who attend seminars may already be at a higher risk of dropping out. This suggests that the observed relationship is likely due to selection bias.

The independent variables in the dropout model were evaluated to determine their contribution to predicting student dropout. Multiple model specifications were examined by testing different combinations of variables and comparing their statistical significance and overall predictive performance. Variables with p-values greater than the 10% significance level were considered not meaningful for predictors and candidates for removal.

From the full model, high school GPA, units enrolled, commuter student status, and seminars attended were consistently found to be statistically significant predictors. High school GPA had a strong negative effect on dropout ($p < .001$), indicating that students with higher academic preparedness are significantly less likely to leave college. Unit enrollment had a positive effect ($p < .001$), showing that students taking more units are more likely to drop out. Commuter status was also significant ($p = .003$), suggesting that commuting influences retention outcomes. Seminar attendance was statistically significant ($p = .003$) and positively associated with dropout probability.

In contrast, age, gender, and registered status were not statistically significant across model specifications, with p-values greater than 0.10 (Age $p = .705$, Gender $p = .691$, Registered status $p = .829$). These variables were tested in combination with other predictors but did not improve

model performance or explanatory power. Additionally, removing these variables did not reduce overall model accuracy, which remained approximately 80.7–80.8%. Therefore, these variables were excluded from the revised model.

The revised logistic regression model includes only the significant predictors and is expressed as:

$$=1.206-1.621(\text{high school GPA})+0.147(\text{units enrolled})-0.800(\text{commuter student status})+0.673(\text{seminars attended})$$

The revised model is preferred because it is simpler while maintaining the same predictive performance as the full model. The removal of insignificant variables is supported by statistical evidence (p-values) and by the fact that model accuracy and explanatory power remained unchanged. This confirms that the excluded variables do not contribute meaningfully to predicting student dropouts.

Revised Model and Validation

After a thorough evaluation of the entire dropout model, a refined logistic regression was created, which exclusively included the four statistically significant predictive factors: high school GPA, units enrolled, commuter status, and seminar attendance. The variables of age, gender, and registration status were not included in the study because their respective p-values exceeded the 10% significance threshold (Age $p = .705$).

The revised model was then validated using an MLP neural network with the same four predictors and a 70/30 split. The network, employing a single hidden layer with three units, attained a training accuracy of 80.1% and a testing accuracy of 80.8%. Demonstrating results virtually indistinguishable from those of logistic regression. The AUC value achieved was 0.750, indicating a moderate enhancement compared to the logistic model. Significantly, the MLP identified roughly twice the number of at-risk students compared to the logistic model on the independent test set, thus serving as a valuable supplementary screening instrument. Therefore, the proposed methodology is to utilize the logistic equation for interpretation and probability estimation, and to employ the MLP as a complementary tool for flagging additional at-risk students who fall beneath the logistic model's classification threshold.

Probability Predictions

When the revised logistic regression model is applied, with GPA maintained at the sample average and commuter status fixed as non-commuter, the estimated probabilities of dropout exhibit a significant escalation as the number of units enrolled increases from 3 to 18. At three

units, the projected probability of attrition is roughly 4.5% for non-seminar participants and 8.4% for seminar participants. With a median enrollment of 12 units, the figures are observed to ascend to roughly 14.9% and 25.6%, correspondingly. At 18 units, the estimated probabilities are approximately 29.8% and 45.4%. The presented statistics compellingly suggest that academic workload represents a potent and malleable factor influencing student attrition. Moreover, the disparity observed between seminar and non-seminar student attrition rates at each academic tier is indicative of a pre-existing selection bias, rather than a consequential effect of the seminar itself.

When units are held at the sample, mean and commuter status is set to non-commuter, the estimated dropout probabilities, as a function of high school GPA, display a more pronounced gradient. Students who possess a GPA of 4.0 have an estimated dropout probability of roughly 3.1% when they do not attend a seminar, and 5.9% when they do. When the GPA is 2.0, the likelihoods elevate to approximately 44.8% and 61.4%. When the GPA range is at its lowest, near 1.0, the estimated probabilities are approximately 80.4% and 88.9%. The forecasts emphasize that preceding academic readiness stands as the paramount indicator of retention available to the university. Further analysis through cross-tabulation of GPA by units enrolled, while controlling commuter status and seminar attendance at zero, indicates a distinct risk stratification. This gradient ranges from students with low GPAs and high unit enrollment, who are in the highest-risk category, to those with high GPAs and low unit enrollment, who fall into the lowest-risk category. Consequently, this naturally delineates the population requiring early intervention.

Conclusions and Recommendations

The analysis demonstrates that student retention is primarily influenced by high school GPA, units enrolled, and commuter status. Students with lower academic preparation and heavier course loads are more likely to drop out after their first year. The college administration should consider implementing alert systems that reach students with the highest risk before the semester begins.

The current data does not clearly define whether seminars improve retention. The positive association between seminar attendance and dropout reflects selection bias as at-risk students attend. The college should consider random assignments in the coming year, which eliminates selection bias and produces a clean estimate on retention. If full randomization is not feasible, comparing outcomes among students with similar GPA and course loads would provide a reasonable approximation.

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Several broader initiatives could support retention. Course load monitoring for incoming first-year students is encouraged, as dropout risk approximately triples when moving from 9 to 18 units. Expanding financial aid, strengthening mental health services, and increasing access to academic support resources such as tutoring centers would all provide help. Conducting regular surveys would provide insight for students and administration can adapt support strategies. With thoughtful targeting and rigorous evaluation, the Student Support Office is well-positioned to reduce first-year attrition at Mont St. Michel College.

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